Enterprise Cloud-Native Deployment Topology: FastAPI Microservices, Kubernetes & WORM Vault

INGRESS & SECURITY TIER
TLS 1.3 & API Gateway

Core Banking System (CBS)

Temenos T24 / Finacle / Flexcube

ISO 20022 clearing webhooks

Nginx / Envoy Ingress

mTLS 1.3, Rate-Limit, WAF

Path sanitization & anti-CSRF

JWT Authentication & RBAC

Maker, Checker, Compliance Auditor

Four-Eyes Dual Sign-Off gate

Out: Authenticated REST/WS Streams

Zero-Trust Banking Boundary

FASTAPI WORKER POOL
Asynchronous Orchestration

FastAPI Async Engine

Uvicorn ASGI Workers, Python 3.12

Sub-millisecond routing loop

Fast-Path LRU Subgraph L1

Capacity: 100k, 0.45 ms Hit

Distributed Redis L2 Cache

3-Tier Conformal Arbiter

Zero-Divergence Operational Triage

Instant Tier-3 STP Auto-Clear

Out: Triage Verdict: T1 / T2 / T3

High-Throughput Microservice Tier

INFERENCE & AGENT SWARM
C-STGB Pods & 4 AI Detectives

C-STGB Neural Pods

Triton / TorchScript Inference

Zero-GPU commodity CPU native

Autonomous 4-Agent Swarm

Investigator → Drafter → Auditor

Sub-30s regulatory dossier

Compiler AST Grammar

$\mathcal{V}(\mathcal{T}) \subseteq \mathcal{V}_{\text{sub}}$ (Prop 6.1)

0% AI entity hallucination

Out: Sealed SAR/STR Regulatory Dossier

Deterministic AI Investigation

GOVERNANCE & CLIENT TIER
WORM Ledger & React 18 UI

Cryptographic WORM Vault

SHA-256 Merkle Receipts (FRE 902(11))

Immutable tamper-evident ledger

PostgreSQL Cold Archival

Case Management & Audit Trail

Statutory SLA countdown docket

React 18 Command Center

Cytoscape 2D + Three.js 3D WebGL

Real-time WebSocket telemetry

Out: Court-Admissible Evidentiary Record

Federal Reserve SR 26-2 Aligned